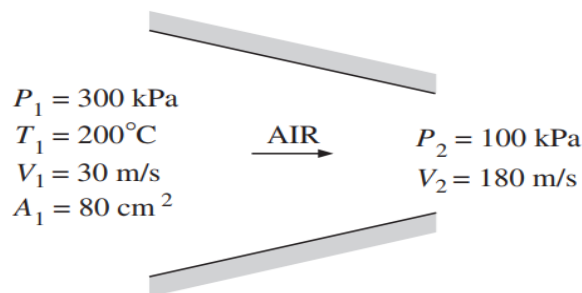


	BAHIR DAR INSTITUTE OF TECHNOLOGY (BiT) FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING 2nd year Mechanical Engineering students Engineering Thermodynamics I Group assignment (13%) Submission date: 24/01/2017 E.C	
---	---	---

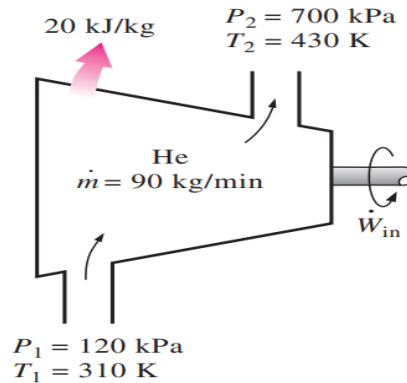
1. What is the purpose of introducing the second law of thermodynamics? Does the process occur when it only satisfying the second law of thermodynamics? Discuss comparing to first law of thermodynamics.
2. In choosing between an electrical heater and heat pump for heating the room, which one is preferable in terms of energy efficiency and environmental impact? State your reason clearly.
3. Prove the second Carnot principle which states “The efficiencies of all reversible heat engines operating between the same two reservoirs are the same”.
4. Can we save all the heat rejected (Q_{out}) in a steam power plant? Discuss with diagram
5. What is the difference between a refrigerator and a heat pump? Support your answer with diagram.
6. Why are engineers interested in reversible processes? Explain.
7. Show that the ENTROPY (S) is state function(property).
8. Air enters an adiabatic nozzle steadily at 300 kPa, 200°C, and 30 m/s and leaves at 100 kPa and 180 m/s. The inlet area of the nozzle is 80 cm². Determine (a) the mass flow rate through the nozzle, (b) the exit temperature of the air, and (c) the exit area of the nozzle.



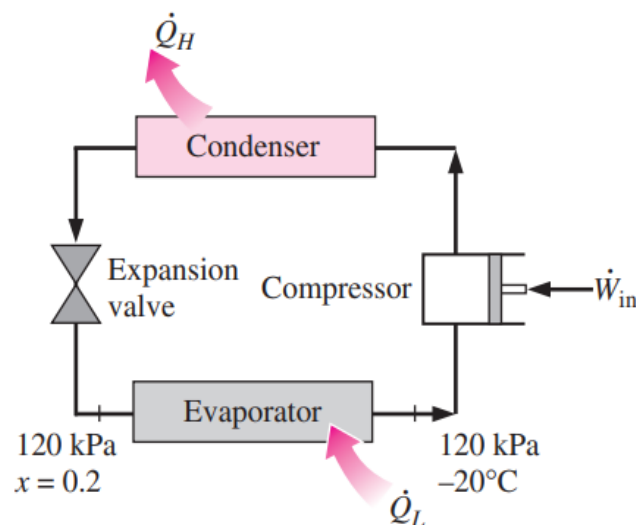
9. Helium is to be compressed from 120 kPa and 310 K to 700 kPa and 430 K. A heat loss of 20 kJ/kg occurs during the compression process. Neglecting kinetic energy changes, determine the power input required for a mass flow rate of 90 kg/min.



BAHIR DAR INSTITUTE OF TECHNOLOGY (BiT)
FACULTY OF MECHANICAL AND INDUSTRIAL
ENGINEERING
2nd year Mechanical Engineering students
Engineering Thermodynamics I
Group assignment (13%)
Submission date: 24/01/2017 E.C



10. Carbon dioxide enters an adiabatic compressor at 100 kPa and 300 K at a rate of 0.5 kg/s and leaves at 600 kPa and 450 K. Neglecting kinetic energy changes, determine (a) the volume flow rate of the carbon dioxide at the compressor inlet and (b) the power input to the compressor.
11. Refrigerant-134a enters the evaporator coils placed at the back of the freezer section of a household refrigerator at 120 kPa with a quality of 20 percent and leaves at 120 kPa and -20°C. If the compressor consumes 450 W of power and the COP the refrigerator is 1.2, determine (a) the mass flow rate of the refrigerant and (b) the rate of heat rejected to the kitchen air.





12. A heat pump is used to maintain a house at 22°C by extracting heat from the outside air on a day when the outside air temperature is 2°C . The house is estimated to lose heat at a rate of $110,000 \text{ kJ/h}$, and the heat pump consumes 5 kW of electric power when running. Is this heat pump powerful enough to do the job?

